MTH636 – Game Theory

Lecture 27: Mixed & Behavioral Strategies; SPE Theorem

1 Strategies in Extensive Form Games

Recall that in an extensive form game Γ each player i has a collection of information sets H_i. A *pure strategy* for player i assigns an action to every information set. We now extend this to mixed and behavioral strategies.

Definition: Pure, Mixed, and Behavioral Strategies

Let H_i be the set of information sets for player i, and let $A(h)$ denote the set of actions available at information set h.

- **Pure strategy:** $s_i : H_i \to \bigcup_{h \in H_i} A(h)$ such that $s_i(h) \in A(h)$ for all $h \in H_i$.
- **Behavioral strategy:** $b_i : H_i \to \bigcup_{h \in H_i} \Delta A(h)$ such that $b_i(h) \in \Delta A(h)$ for all $h \in H_i$.
  (A behavioral strategy assigns an *independent* probability distribution over actions at *each* information set.)
- **Mixed strategy (over pure strategies):** $\sigma_i : S_i \to [0,1]$ such that $\sum_{s_i \in S_i} \sigma_i(s_i) = 1$.
  (A mixed strategy is a probability distribution over the set of *all* pure strategies S_i.)

Note

A **mixed strategy** randomises over complete plans of action (pure strategies), while a **behavioral strategy** randomises independently at each information set. In games of *perfect recall*, the two concepts are equivalent (Kuhn's Theorem).

Example – Mixed vs. Behavioral Strategies

Consider a game where Player I moves at the root n_0 (actions A and B) and then again inside an information set $\{n_1, n_3\}$ (actions ℓ and r), and Player II has an information set with actions L and R.

Suppose Player I's *mixed strategy* puts weight:

$$\sigma_1(Al) = 0.5, \qquad \sigma_1(Ar) = 0.1,$$
$$\sigma_1(Bl) = 0.3, \qquad \sigma_1(Br) = 0.1.$$

(These are the probabilities on the four pure strategies Al, Ar, Bl, Br.)

The corresponding *behavioral strategy* b_1 is read off by conditioning:

- At the root: $\Pr[\text{play } A] = 0.5 + 0.1 = 0.6, \quad \Pr[\text{play } B] = 0.3 + 0.1 = 0.4.$
- Inside the information set $\{n_1, n_3\}$: $b_1'(\ell) = 0.5$, $b_1'(r) = 0.5$.

Given a mixed strategy σ_i, its **induced behavioral strategy** $\bar{\sigma}_i$ is defined as follows. Let $\bar{\sigma}_i(h)$ be:

- σ_i outside Γ_n (i.e. inherited from the mixed strategy in the rest of the game), and
- $\bar{\sigma}_i$ inside Γ_n (re-normalised conditional on reaching h).

Formally, if $\sigma_i(A) = \sigma_i(A) \cdot \Gamma_n$-weight, then

$$\bar{\sigma}_i(A) = \sigma_i(A), \quad \bar{\sigma}_i(B) = \sigma_i(B), \quad \bar{\sigma}_i(\ell) = \frac{\sigma_i(\ell)}{\sigma_i(\ell) + \sigma_i(r)}, \quad \bar{\sigma}_i(r) = \frac{\sigma_i(r)}{\sigma_i(\ell) + \sigma_i(r)}.$$

2 Subgame Perfect Equilibrium – Main Theorem

Theorem: SPE and Reached Subgames (Theorem)

Let σ^* be a Nash Equilibrium (either in mixed strategies or in behavioral strategies) of a game Γ, and let Γ_n be a subgame of the game. If $\rho(n; \sigma^*) > 0$ (i.e. node n is reached with positive probability under σ^*), then σ^* is also a Nash Equilibrium for the subgame Γ_n.

Proof. Suppose σ restricted to Γ_n is *not* an NE of Γ_n. Then there exists a player i and a strategy σ_i' in the subgame Γ_n such that σ_i' gives player i strictly more utility in Γ_n.

Construct a new strategy for player i in the full game:

$$\bar{\sigma}_i = \begin{cases} \sigma_i & \text{outside } \Gamma_n, \\ \sigma_i' & \text{inside } \Gamma_n. \end{cases}$$

Since $\rho(n; \sigma^*) > 0$, we have $\rho\Big(n; (\bar{\sigma}_i, \sigma_{-i})\Big) > 0$. From here one can show that

$$u_i(\sigma^*) < u_i\Big(\bar{\sigma}_i, \sigma_{-i}\Big),$$

which contradicts the assumption that σ^* is an NE of Γ. Therefore σ^* restricted to Γ_n must be an NE of Γ_n.

(The full proof can be found in Solan's book.) □

3 Corollary and Remarks

Proposition: Corollary – Sufficient Condition for SPE

Suppose σ is an NE of a game Γ, and under σ all strategies of all players are given positive probabilities, and all chance nodes give positive probability to all edges coming out of them. Then σ is an SPE.

Note

The corollary says: if *every* subgame is on the equilibrium path (reached with positive probability), then every NE automatically satisfies the SPE condition. This is because the theorem guarantees NE in all reached subgames, and under the corollary's conditions all subgames are reached.

Example Illustrating the Corollary

Consider the following extensive form game. Player I moves at the root (actions A, B, C). After A, Player II moves (information set $\{v_1\}$, actions ℓ and r). After B, Player II moves (information set $\{v_2\}$, actions ℓ and r). The chance node O following C assigns probability $\frac{1}{4}$ to one branch and $\frac{3}{4}$ to another.

Payoffs from left to right (terminal nodes):

$$\begin{aligned}&(4,1),\ (7,2) \quad \text{from } A,\ell \text{ and } A,r,\\ &(6,3),\ (4,0) \quad \text{from } B,\ell \text{ and } B,r,\\ &(0,0),\ (8,4),\ (8,8),\ (0,3) \quad \text{from } C \text{ branches.}\end{aligned}$$

After resolving the chance node at C:

$$\frac{1}{4}(8,4)+\frac{3}{4}(8,8)=(8,7).$$

The reduced game (after replacing chance node with expected payoffs) is:

	ℓ	r
A	(4, 1)	(7, 2)
B	(6, 3)	(4, 0)
C	(8, 7)	(8, 7)

Nash Equilibria: (C,ℓ) and (C,r) are both NE.

Backward Induction Equilibria: $(C_A,\ \ell R)$ and $(C_A,\ rR)$ are the two backward induction equilibria of the game.

Theorem: Remarks

R-1. Every backward induction equilibrium is an SPE.

R-2. For any extensive form game, there always exists an SPE.

Proof: We will discuss this in the next class.

Intuition

Backward induction – working from the terminal nodes backwards – always produces an SPE. Moreover, the existence of an SPE for *any* finite extensive form game (even with imperfect information) is guaranteed.